Project 2
Final Presentation

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Research Questions

- Q1 - Does peripheral inflammation predict one-year decline in episodic memory and AD-signature cortical thickness?
  - 1a - Higher baseline cytokine and chemokine levels will predict greater declines in memory and cortical thickness.
  - 1b - Greater increases in cytokine and chemokine levels over one year will be associated with greater declines in memory and cortical thickness.
- Q2 - Does amyloid deposition interact with inflammation and affect the decline of episodic memory and AD-signature cortical thickness?
  - 1a - Higher baseline cytokine and chemokine levels and amyloid deposition will predict greater declines in memory and cortical thickness than either predictor alone.

Data Analysis Plan

Similar Model Structures

- Outcomes - Changes in CVLT-20' Delay, CVLT-Recognition, Benson Figure, MST, Story Recall, & Cortical Thickness
- Primary Predictors - 4 inflammation markers: IL-6, TNF-α, MCP-1, & Eotaxin-1
- Covariates - Outcomes at Baseline, Age, Sex, APOE genotype (plan to evaluate others)

Aim 1

- 1a - predictor at baseline
  1b - change in predictors
- Assessing the association of the individual inflammatory markers and the changes in outcomes

Aim 2

- Add interaction term - amyloid deposition and inflammation markers
- Add covariate for global SUVR
- Assessing the significance of the interaction term

Power Analysis - Aim 1

- Goal: Detect correlation different from 0
- Sample Size = 175
- Effect Sizes = 0.2 to 0.7 by 0.05
- Alpha Levels = 0.05/4, 0.05/6, 0.05/24
- Used corr.1samp

- Power of at least 80% achieved across all α levels for correlations greater than 0.30
- The analysis is powered to detect moderate correlations

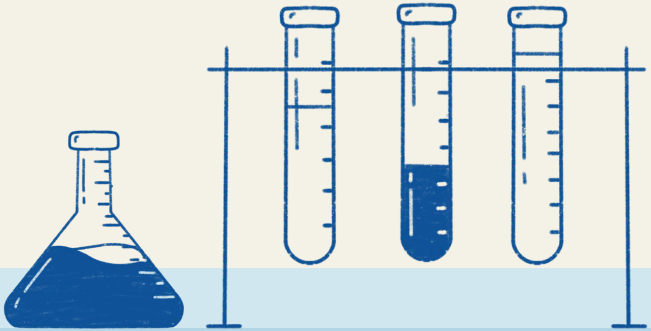
Aim 1: Statistical Power by Effect Size and Significance Level (N = 175)

	Significance Level		
r	α = 0.05/4	α = 0.05/6	α = 0.05/24
0.20	56.7%	51.1%	34%
0.25	80.6%	76.5%	61.1%
0.30	94.2%	92.4%	84%
0.35	99%	98.5%	95.8%
0.40	99.9%	99.8%	99.4%
0.45	100%	100%	100%
0.50	100%	100%	100%
0.55	100%	100%	100%
0.60	100%	100%	100%
0.65	100%	100%	100%
0.70	100%	100%	100%

Power Analysis - Aim 2

Aim 2: Minimum detectable difference in correlations at 80% power by amyloid group size and significance level (N = 175)

Group Size (N+/N-)	Significance Level	Min Detectable r Difference
56/25	α = 0.05/4	0.40
56/25	α = 0.05/6	0.45
56/25	α = 0.05/24	0.50
88/25	α = 0.05/4	0.40
88/25	α = 0.05/6	0.40
88/25	α = 0.05/24	0.45
120/25	α = 0.05/4	0.40
120/25	α = 0.05/6	0.45
120/25	α = 0.05/24	0.50

- Goal: Detect difference in correlations between amyloid groups
- Amyloid Positive Sample Sizes = 25, 56, 88, 120, 150
- Effect Sizes = 0.2 to 0.7 by 0.05
- Alpha Levels = 0.05/4, 0.05/6, 0.05/24
- Used corr.2samp
- Power did not reach 80% for the smallest and largest amyloid-positive group sizes examined. For the remaining group size scenarios, the minimum detectable effect sizes were ~0.4–0.5 across α levels
- The analysis is powered to detect moderate to large difference in correlations between amyloid groups

Questions?